8.3 Proving Triangle Similarity by SSS and SAS

Name _____ Period ____ Date _____

Geometry 8.3 (30-minute plan) · Student sheet

I can…

- compare the ratios of corresponding sides to decide whether two triangles are similar (SSS).
- use an included angle and two proportional sides to show two triangles are similar (SAS).
- find the value of a variable that makes two triangles similar.

Vocabulary · fill in the missing word

SSS Similarity Theorem — if the _____ side lengths of two triangles are proportional, the triangles are similar

SAS Similarity Theorem — if an angle of one triangle is congruent to an angle of another and the sides including the angles are _____, the triangles are similar

Included angle — the angle _____ by two sides

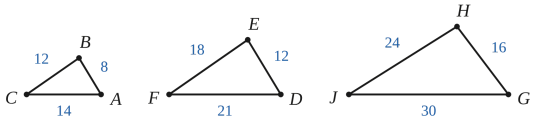
Part 1 · Notes and examples (fill in each note, then work the example)

SSS Similarity Theorem: if the corresponding _____ _____ of two triangles are proportional, then the triangles are similar.

Checking SSS: Compare shortest to shortest, _____ to _____, then the remaining pair. All the ratios must be _____.

Example 1

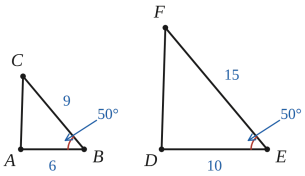
Which of the triangles are similar? △ABC has sides 8, 12, 14. △DEF has sides 12, 18, 21. △GHJ has sides 16, 24, 30.



SAS Similarity Theorem: if an angle of one triangle is _____ to an angle of another and the lengths of the sides _____ these angles are proportional, then the triangles are similar.

Example 2

△ABC has m∠B = 50°, AB = 6, and BC = 9. △DEF has m∠E = 50°, DE = 10, and EF = 15. Are the triangles similar? Explain.



Example 3

△PQR has PQ = 6, QR = 8, and m∠R = 40°. △STU has ST = 9, TU = 12, and m∠U = 40°. Can you conclude that the triangles are similar?

Part 2 · Partner practice

Partner practice

In △ABC and △DEF, ∠A ≅ ∠D = 72°, AB = 8, AC = 12, DE = 6, and DF = 9. Are the triangles similar? Explain.